HopLens: Interpretable Path Scoring for Multi-Hop Retrieval

Hana Kobayashi (1), Mateo Rossi (1)
(1) Northern Isles University
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Abstract

Multi-hop retrievers rarely explain why a chain of passages was chosen. HopLens scores each candidate chain with an interpretable path-aware reranking module that assigns a weight to every hop transition. HopLens reaches 56.2 and 44.8 EM on KaveriHop 2-hop and 3-hop, and 39.5 EM on TriHop-Bench 3-hop. HopLens achieves state-of-the-art exact match on 3-hop questions in both KaveriHop and TriHop-Bench.

Keywords: interpretability, reranking, multi-hop retrieval

1 Introduction

Dense multi-hop retrievers such as DenseHop [1] and graph walkers such as GraphWalker-QA [2] produce chains without explanation. We propose to rerank chains with a transparent per-hop scorer.

2 Method

HopLens retrieves 50 candidate chains with a dense retriever initialised from dense passage retrieval [5], then applies path-aware reranking: a cross-encoder scores each transition between consecutive passages, and the chain score is a weighted sum of transition scores. The transition weights are shown to the user as an explanation.

3 Experiments

We evaluate on KaveriHop [3] and TriHop-Bench [4], comparing with DenseHop [1] and GraphWalker-QA [2] using published numbers.

Method	Source	KaveriHop 2-hop EM	KaveriHop 3-hop EM	TriHop 3-hop EM
DenseHop	[3]	48.2	31.5	–
GraphWalker-QA	[2]	50.1	39.8	–
GraphWalker-QA	[4]	–	–	34.7
HopLens	ours	56.2	44.8	39.5

Table 1: Exact match on KaveriHop and TriHop-Bench.

4 Conclusion

HopLens achieves state-of-the-art exact match on 3-hop questions in both KaveriHop and TriHop-Bench while explaining its choices.

References

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